

普物公式

多普勒效应: $\frac{f'}{f} = \frac{v}{v \pm v_{\text{com}}}$



相邻波节之间具有相同的相位, 波节两侧的振动相位相反, 即相位差 π .

$$t'_1 = \frac{t_1 - \frac{v}{c^2} x_1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$t'_2 = \frac{t_2 - \frac{v}{c^2} x_2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$x'_1 = \frac{x_1 - vt_1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$x'_2 = \frac{x_2 - vt_2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M_m}}$$

$$\bar{v} = \sqrt{\frac{8RT}{\pi M_m}}$$

$$v_p = \sqrt{\frac{2RT}{M_m}}$$

$$\bar{z} = \sqrt{2} n \bar{v} \delta \quad \bar{\lambda} = \frac{\bar{v} t}{\bar{z}} = \frac{\bar{v}}{\bar{z}} = \frac{1}{\sqrt{2} n \delta}$$

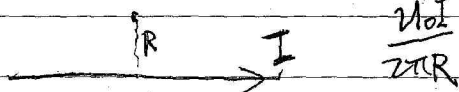
碰撞频率

平均自由程

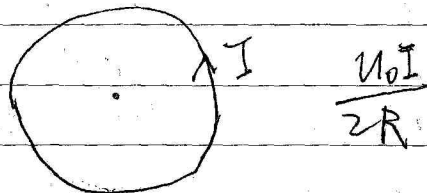
$$ds = \frac{dQ}{T} = \frac{dU + PdV}{T}$$

$$dU = v \cdot C_{v,m} dT$$

$$C_{v,m} = \left(\frac{\partial U_m}{\partial T} \right)_v \quad C_{p,m} = \left(\frac{\partial H_m}{\partial T} \right)_p$$



可见光: 390 ~ 780 nm



干涉亮: $d \sin \theta = k\lambda$

光栅分辨率本领: $R = \frac{\lambda}{\Delta \lambda} = kN$

衍射暗: $a \sin \theta = k'\lambda$

$\delta \theta = \theta_{\min} = 1.22 \cdot \frac{\lambda}{D}$ (最小分辨角即角分辨率)

分辨率 $R = \frac{1}{\delta \theta}$

